

100-GHz and >0.5-A/W Balanced Photodiodes with Interference-Enhanced Absorption in O- and C-Bands

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Abstract We demonstrate vertical-illumination InGaAs/InP balanced photodiodes (PDs) using interference-enhanced absorption for high-speed coherent detection. O-band- and C-band-optimized PDs achieve 100 GHz bandwidth, 0.52 A/W responsivity, 0.06 nA dark current, <0.1 dB polarization-dependent loss (PDL), and <-35 dB common-mode rejection ratio (CMRR). ©2026 The Author(s)

Introduction

The growth of data traffic driven by generative artificial intelligence (AI) is increasing the demand for higher-symbol-rate systems. Receivers for 3.2 Tbit Ethernet will need photodiodes (PDs) with bandwidths of 100 GHz for symbol rates beyond 200 GBaud. Short-reach systems have relied on intensity modulation and direct detection (IMDD), but their transmission reach shortens as the symbol rate rises. For that reason, digital coherent transmission is now being considered even for shorter links [1]. In coherent receivers designed for low power consumption, responsivity is one of the key parameters because it directly affects the required local-oscillator power and optical amplifier gain [2,3]. Since coherent detection generally relies on differential detection, balanced PDs are key components in such receivers. In addition to bandwidth and responsivity, low dark current is beneficial for long-term device reliability. Low polarization-dependent loss (PDL) is also valuable because polarization fluctuation can cause responsivity imbalance and unstable differential output; therefore, reducing the PDL can relax polarization-control requirements. Table 1 compares our work with recently reported balanced PDs [4–8].

A balanced PD consists of two PDs, PD_{NEGA} and PD_{POS}, and generates a differential output. The challenge is that a balanced PD has the parallel capacitance of both PDs, which inherently limits bandwidth, and reducing the active area to increase bandwidth compromises responsivity and reliability. Furthermore, precise fabrication techniques are required to match the responsivity and RF characteristics between the two PDs, because any mismatch increases residual noise and degrades the common-mode rejection ratio (CMRR).

In this work, we present a vertical-illumination balanced PD with interference-based absorption enhancement designed for high bandwidth and

high responsivity. Our PD achieved a high responsivity of 0.52 A/W in the O- and C-bands while maintaining the 100 GHz bandwidth required for 200 GBaud-class operation. It also exhibited a low dark current of 0.06 nA, PDL below 0.1 dB, and a calculated CMRR below -35 dB. Overall, our balanced PD combines high speed, high responsivity, excellent CMRR, and low polarization dependence.

Device Structure and Design

Figure 1 shows the conceptual structure of the proposed balanced PD. The cathode of PD_{NEGA} and the anode of PD_{POS} are connected to the signal line to obtain a differential output. Two monolithically integrated metal–insulator–metal (MIM)

Tab. 2: Performance comparison of this work and recently reported high-speed balanced PDs

Ref.	BW [GHz]	Resp. [A/W]	Id [nA]	PDL [dB]	CMRR [dB]
This	100	0.52	0.06	<0.1	<-35
[4]	150	0.1	NA	NA	<-30
[5]	100	0.25	5	0.75	NA
[6]	100	0.5	<100	NA	<-20
[7]	80	0.55	3	<0.5	<-20
[8]	30	0.72	<100	NA	<-26

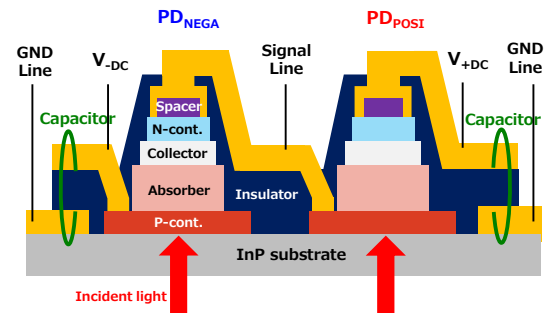


Fig. 1: Schematic cross section of the proposed balanced PD. PD_{NEGA} and PD_{POS} are integrated in parallel, and the differential signal is extracted through the signal line with two integrated capacitors.

capacitors provide RF coupling between the bias lines and ground. This design removes the need for an external bias-tee that may degrade bandwidth. Note that the measured frequency response reflects not only the PDs themselves but also the integrated capacitors and wiring. To suppress the mismatch between the two PDs which can degrade CMRR, a symmetrical layout was adopted with tighter fabrication control.

Both PD_{NEGA} and PD_{POSI} are 6- μm -diameter vertical-illumination InGaAs/InP PDs, grown on the same substrate and based on an inverted p-down structure. This semiconductor structure is based on our previously established design of a single PD [9]. The inverted p-down structure suppresses the electric field near the sidewall and reduces surface-related leakage. In single-PD devices using the same semiconductor platform, we have already demonstrated reliability that satisfies Telcordia GR-468-CORE [10]. Another advantage of the vertical-illumination structure is its lower polarization dependence than that of waveguide PDs, which stabilizes responsivity against polarization fluctuation.

High responsivity was achieved by interference-based absorption enhancement [8]. While a thicker absorber increases responsivity, it also increases carrier transit time and limits bandwidth. In this design, the absorber was kept thin and the absorption was enhanced by multilayer interference, enabling high responsivity without sacrificing bandwidth. A key design point is control of the resonant wavelength through the SiN spacer thickness. The resonant wavelength is determined by the optical cavity condition in the multilayer structure and shifts toward longer wavelengths as the SiN thickness increases. This allows the target wavelength to be tuned during processing without changing the epitaxial wafer. In practice, a SiN film with the desired thickness is deposited on the wafer surface and patterned by dry etching. Repeating this process enables devices with different SiN spacer thicknesses, and hence different resonant wavelengths, to be fabricated with high precision on the same epitaxial wafer. As shown in Fig. 2, varying the SiN thickness from 0 to 400 nm shifts the resonant wavelength from about 1310 to 1670 nm. Accordingly, the O-band device was designed without a SiN spacer, whereas the C-band device employed a 270 nm spacer to maximize responsivity near 1550 nm, allowing both devices to be fabricated on the same platform in a single process flow.

Static Characteristics

Figure 3 shows the I-V characteristics of PD_{NEGA} and PD_{POSI} measured individually. Both devices exhibited dark currents of approximately 0.06 nA at the standard operating bias of 3 V, including

leakage current associated with the capacitors and substrate. The dark current is extremely low: more than two orders of magnitude smaller than those of recently reported balanced PDs [4–8]. The difference between the two PDs is also small, which indicates that the fabrication process provides good reproducibility. This result suggests that our balanced-PD structure is favorable for long-term reliable operation.

Figure 4 shows the wavelength dependence

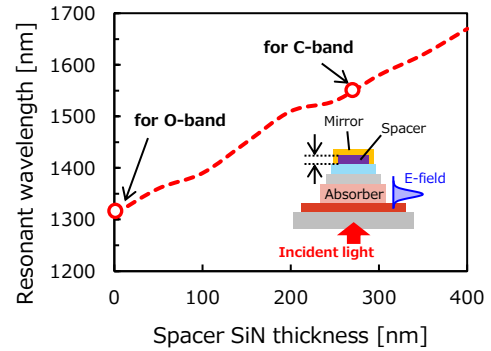


Fig. 2: Calculated resonant wavelength as a function of SiN spacer thickness. The resonant wavelength shifts as the SiN thickness increases, enabling wavelength-selective design on the same epitaxial wafer.

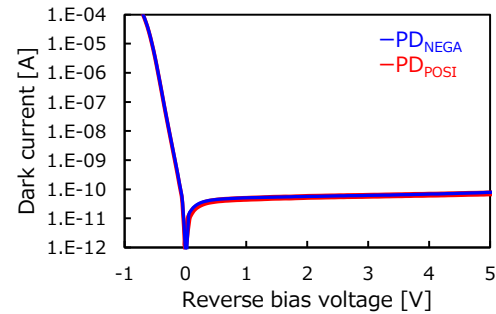


Fig. 3: Measured I-V characteristics of PD_{NEGA} and PD_{POSI}

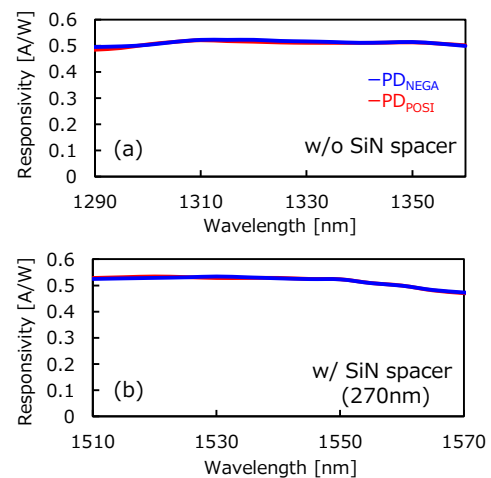


Fig. 4: Wavelength dependence of the responsivity in (a) O-band and (b) C-band.

of responsivity in the O- and C-bands. The balanced PDs fabricated without and with SiN spacers achieved responsivities of 0.52 A/W at 1310 and 1550 nm, respectively. These results are consistent with the design based on SiN-thickness-controlled resonance.

Figure 5 shows the polarization dependence of the normalized responsivity. The measured PDL was below 0.1 dB, although part of this residual value may originate from the measurement setup. This is a clear benefit of the vertical-illumination structure and particularly useful in coherent detection, where polarization fluctuations lead to responsivity variation and unstable differential detection.

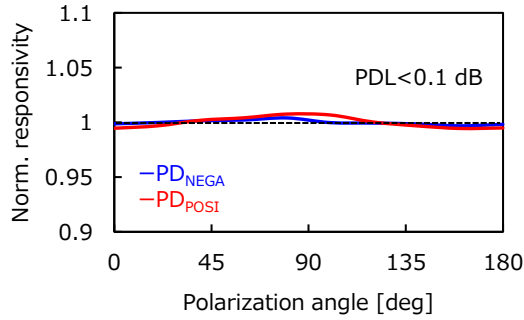


Fig. 5: Normalized responsivity as a function of polarization angle.

High-speed Performance

Figure 6 shows the optical-to-electric (OE) responses of PD_{NEGA} and PD_{POSI} measured at 1310 and 1550 nm using a lightwave component analyzer. The biases were +3 and -3 V for PD_{POSI} and PD_{NEGA}, respectively, and the photocurrent was set to 1 mA. At both wavelengths, the difference between the two PDs was small, and the 3-dB bandwidth was approximately 100 GHz.

Figure 7 shows the CMRR calculated from the measured OE responses using Eq. (1), where I_p

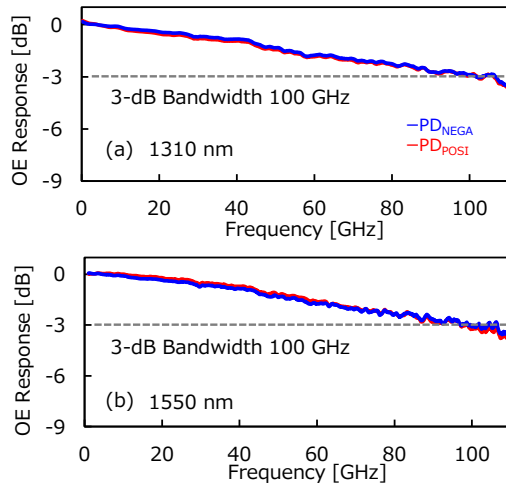


Fig. 6: OE responses of PD_{NEGA} and PD_{POSI} measured at (a) 1310 and (b) 1550 nm.

and I_n are the OE responses of PD_{POSI} and PD_{NEGA}, respectively. The balanced PD exhibited an excellent CMRR below -35 dB up to 100 GHz, demonstrating that it was fabricated successfully as designed.

Although our evaluation is still at the chip level, to the best of our knowledge, this work demonstrates one of the strongest overall combinations of high bandwidth, high responsivity, low dark current, and low PDL together with excellent CMRR.

$$CMRR = 20 \log_{10}[(I_p - I_n)/(I_p + I_n)] \quad (1)$$

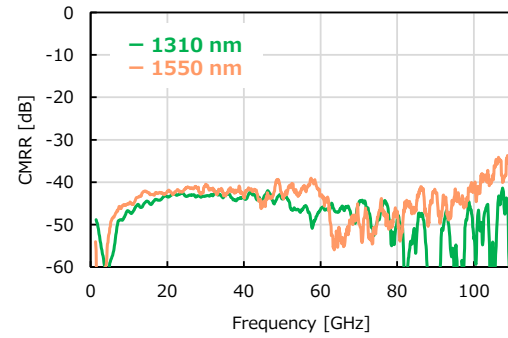


Fig. 7: CMRR calculated from Eq. (1).

Conclusions

We demonstrated a high-speed, high-responsivity vertical-illumination balanced PD with integrated capacitors enabled by interference-based absorption enhancement. The O-band- and C-band-optimized PDs achieved a 3-dB bandwidth of 100 GHz, a responsivity above 0.5 A/W, a dark current of 0.06 nA, low polarization dependence, and CMRR below -35 dB. These results demonstrate that the proposed balanced PD is promising for coherent receivers, including 200 GBaud-class transceivers that require both low power consumption and stable operation.

Acknowledgements

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